

Web Programming

C333-CS614

Dr Walid M. Aly

Lecture#13

jQuery

Reference: <http://www.w3schools.com/jquery/default.asp>

First Example on jQuery

```
<!DOCTYPE html>
<html>
<head>
<script src="jquery-2.1.3.js"></script>
<script>
$(document).ready(function(){
  $("p").click(function(){
    $(this).hide();
  });
});
</script>
</head>
<body>
<p>If you click on me, I will disappear.</p>
<p>Click me away!</p>
<p>Click me too!</p>
</body>
</html>
```

If you click on me, I will disappear.

Click me away!

Click me too!

[jQuery1.html](#)

What is jQuery?

- jQuery is a lightweight, "write less, do more", **JavaScript library**.
- jQuery takes a lot of common tasks that require many lines of JavaScript code to accomplish, and wraps them into **methods** that you can call with **a single line of code**.
- The jQuery library contains the following features:
 - HTML/DOM manipulation
 - CSS manipulation
 - HTML event methods
 - Effects and animations
 - AJAX

jQuery Install

- The jQuery library is a [single JavaScript file](#), and you reference it with the HTML `<script>` tag inside the `<head>` section.
- There are several ways to start using jQuery on your web site. You can:
 - Download the jQuery library from jquery.com

```
<head>  
<script src="jquery-1.11.1.min.js"></script>  
</head>
```

Or

- Include jQuery from a CDN(Content Delivery Network), like Google

```
<script  
src="http://ajax.googleapis.com/ajax/libs/jquery/1.11.1/jquery.min.js">  
</script>
```

jQuery Syntax

- The jQuery syntax is tailor made for **selecting** HTML elements and performing some **action** on the element(s).
- Basic syntax is: **`$(selector).action()`**
 - A \$ sign to define/access jQuery
 - A (*selector*) to "query (or find)" HTML elements
 - A jQuery *action()* to be performed on the element(s)

Examples:

`$(this).hide()` - hides the current element.

`$("p").hide()` - hides all `<p>` elements.

`$(".test").hide()` - hides all elements with `class="test"`.

`$("#test").hide()` - hides the element with `id="test"`.

Document ready event

- It is good practice to wait for the document to be fully loaded and ready before working
- This is to prevent any jQuery code from running before the document is finished loading (is ready)

```
$(document).ready(function(){  
  
    // jQuery methods go here...  
  
});
```

```
$(function(){  
  
    // jQuery methods go here...  
  
});
```

jQuery Selectors

- jQuery selectors allow you to select and manipulate HTML element(s).
- jQuery uses CSS syntax for selector in addition to its own selectors
- All selectors in jQuery start with the dollar sign and parentheses: \$().
- Selectors can be:
 - element Selector
 - #id Selector
 - .class Selector

```
$("p").hide();
```

all <p> elements will be hidden

```
$("#test").hide();
```

the element with id="test" will be hidden

```
$(".test").hide();
```

the elements with class="test" will be hidden

Example#2

```
<!DOCTYPE html> <html>
<head>
<script src="jquery-2.1.3.js"></script>
<script>
$(document).ready(function(){
    $("button#bt1").click(function(){
        $("#test").hide();
    });
});
</script>
</head>
<body>
<h2>This is a heading</h2>
<p>This is a paragraph.</p>
<p id="test">This is another paragraph.</p>
<button id="bt1">Click me</button>
<button>Click me too</button>
</body></html>
```

This is a heading

This is a paragraph.

This is another paragraph.

Click me

Click me too

[jQUERY2.html](#)

More jQuery Selectors

Syntax	Description
<code>\$("*")</code>	Selects all elements
<code>\$(this)</code>	Selects the current HTML element
<code>\$("p.intro")</code>	Selects all <code><p></code> elements with <code>class="intro"</code>
<code>\$("p:first")</code>	Selects the first <code><p></code> element
<code>\$("ul li:first")</code>	Selects the first <code></code> element of the first <code></code>
<code>\$("ul li:first-child")</code>	Selects the first <code></code> element of every <code></code>
<code>\$("[href]")</code>	Selects all elements with an <code>href</code> attribute
<code>\$("a[target='_blank']")</code>	Selects all <code><a></code> elements with a <code>target</code> attribute value equal to <code>"_blank"</code>
<code>\$("a[target!='_blank']")</code>	Selects all <code><a></code> elements with a <code>target</code> attribute value NOT equal to <code>"_blank"</code>
<code>\$(":button")</code>	Selects all <code><button></code> elements and <code><input></code> elements of <code>type="button"</code>
<code>\$("tr:even")</code>	Selects all even <code><tr></code> elements
<code>\$("tr:odd")</code>	Selects all odd <code><tr></code> elements

Example#3

```
<!DOCTYPE html>
<html>
<head>
<script src="jquery-2.1.3.js"></script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $(this).hide();
  });
});
</script>
</head>
<body>
<h2>This is a heading</h2>
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
<button>Click me</button>
</body>
</html>
```

This is a heading

This is a paragraph.

This is another paragraph.

Click me

[jQuery3.html](#)

jQuery Event Methods

- **\$(document).ready()**
 - The `$(document).ready()` method allows us to execute a function when the document is fully loaded.
- **click()**
 - The `click()` method attaches an event handler function to an HTML element.
- **dblclick()**
 - The `dblclick()` method attaches an event handler function to an HTML element
- **mouseenter()**
 - The `mouseenter()` method attaches an event handler function to an HTML element.

mousedown()

mouseup()

hover()

focus()

blur()

Example#4

```
<!DOCTYPE html>
<html>
<head>
<script src="jquery-2.1.3.js"></script>
<script>
$(document).ready(function(){
  $("#p1").mouseleave(function(){
    alert("Bye! You now leave p1!");
  });
});
</script>
</head>
<body>

<p id="p1">This is a paragraph.</p>

</body>
</html>
```

[jQUERY4.html](#)

jQuery Effects:

- Hide and Show-

- **Syntax:**
 - `$(selector).hide(speed,callback);`
`$(selector).show(speed,callback);`
 - `$(selector).toggle(speed,callback);`
- The optional speed parameter specifies the speed of the hiding/showing, and can take the following values: "slow", "fast", or milliseconds.
- The optional callback parameter is a function to be executed after the hide() or show() method completes

```
$("#show").click(function(){  
    $("p").show();  
});
```

```
$("#button").click(function(){  
    $("p").toggle();  
});
```

```
$("#button").click(function(){  
    $("p").hide(1000);  
});
```

jQuery Effects

Fading

- With jQuery you can fade an element in and out of visibility.
- jQuery has the following fade methods:
 - `fadeIn()`:fade in a hidden element.
 - `fadeOut()`:fade out a visible element
 - `fadeToggle()`:If the elements are faded out, `fadeToggle()` will fade them in. If the elements are faded in, `fadeToggle()` will fade them out.
 - `fadeTo()`: fading to a given opacity (value between 0 and 1)
lower value makes the element more transparent.
ex: `$("#div1").fadeTo("slow",0.15);`

Example#5



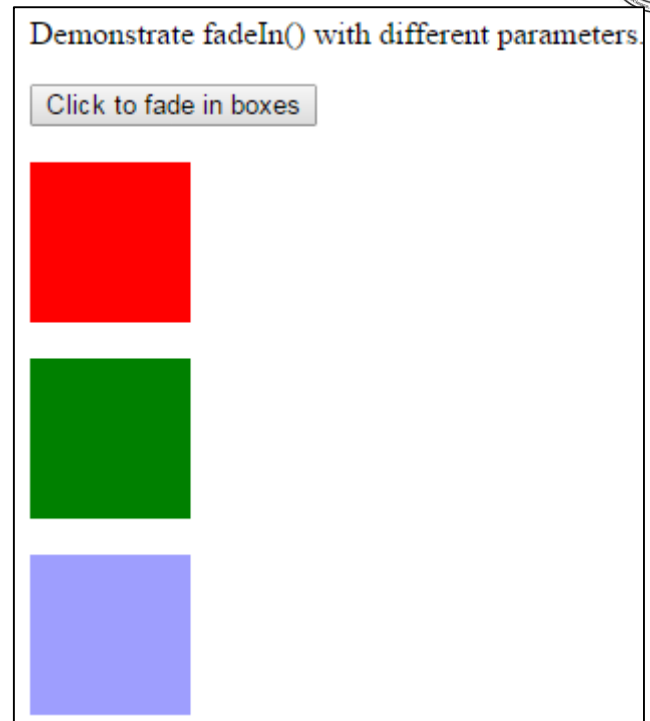
```
<!DOCTYPE html><html> <head>
<script src="jquery-2.1.3.js"></script>
<script>
```

```
$(document).ready(function(){
  $("button").click(function(){
    $("#div1").fadeIn();
    $("#div2").fadeIn("slow");
    $("#div3").fadeIn(6000);
  });
});
```

```
</script>
</head><body>
```

```
<p>Demonstrate fadeIn() with different parameters.</p>
<button>Click to fade in boxes</button>
<br><br>
```

```
<div id="div1" style="width:80px;height:80px;display:none;background-color:red;"></div><br>
<div id="div2" style="width:80px;height:80px;display:none;background-color:green;"></div><br>
<div id="div3" style="width:80px;height:80px;display:none;background-color:blue;"></div>
</body></html>
```



[jQuery5.html](#)

jQuery Effects - Animation

- **Syntax:**

- `$(selector).animate({params},speed,callback);`
- The required params parameter defines the CSS properties to be animated.
- The optional speed parameter specifies the duration of the effect. It can take the following values: "**slow**", "**fast**", or **milliseconds**.
- The optional callback parameter is a function to be executed after the animation completes.

Example#6

```
<!DOCTYPE html><html><head>
<script src="jquery-2.1.3.js"></script>
<script>
$(document).ready(function(){
    $("button").click(function(){
        $("#div").animate({left:'250px'}));
    });
});
</script>
</head><body>
<button>Start Animation</button>
<p>By default, all HTML elements have a static position, and cannot be
moved. To manipulate the position, remember to first set the CSS
position property of the element to relative, fixed, or absolute!</p>
<div
style="background:#98bf21;height:100px;width:100px;position:absolute;">
</div>
</body></html>
```

```
$("#button").click(function(){
    $("#div").animate({
        left:'250px',
        opacity:'0.5',
        height:'150px',
        width:'150px'
    });
});
```

[jQuery6.html](#)

jQuery - css() Method

- the `css()` method sets or returns one or more style properties for the selected elements.
 - Return a CSS Property
 - Syntax: `("propertyname");`
 - `$("p").css("background-color");` // for first paragraph
 - Set a CSS Property
 - Syntax: `css("propertyname","value");`
 - `$("p").css("background-color","yellow");` // for all paragraphs
 - Set Multiple CSS Properties
 - Syntax: `css({"propertyname":"value","propertyname":"value",...});`
 - `$("p").css({"background-color":"yellow","font-size":"200%"});`

```
$("#input").focus(function(){  
    $(this).css("background-color","#cccccc");  
});
```

Example#7

```
<!DOCTYPE html><html><head>
<script src="jquery-2.1.3.js"></script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("#para1").css({"background-color":"yellow","font-size":"200%"});
  });
});
</script>
</head>

<body>
<h2>This is a heading</h2>
<p style="background-color:#ff0000">This is a paragraph.</p>
<p id="para1" style="background-color:#00ff00">This is a paragraph.</p>
<p style="background-color:#0000ff">This is a paragraph.</p>
<p>This is a paragraph.</p>
<button>Set a style for a paragraph</button>
</body>
</html>
```

[jQUERY7.html](#)